

Artificial Intelligence and Productivity in Cognitively Intensive Work: Evidence from a Randomized Controlled Trial

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Abstract

We study whether access to expert-style AI feedback raises the productivity of early-career researchers engaged in cognitively intensive work. In a randomized controlled trial with more than 100 early-career economists, we assign treated participants to a structured, multi-agent AI system that delivers diagnostic feedback, the high-context evaluative work that determines publication success. We measure productivity as the quality of research relative to the effort it requires, using within-person flow items collected at three and six months. We test whether the productivity effect operates through a reallocation of effort toward judgment-intensive tasks, and whether gains concentrate among researchers closer to the knowledge frontier or accrue disproportionately to those still acquiring expertise, a distinction with direct implications for whether AI democratizes or amplifies existing inequality in production.

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Table 1: Project Timeline

Date	Milestone / Activity
2024	Development and piloting of the ResearchAI Development Suite platform
Jan 2026	Finalization of experimental protocol and pre-registration (AEARCTR-0017749)
Feb–Mar 2026	Recruitment window: baseline survey administered to all eligible participants
Apr 2026	Randomization; Treatment group receives immediate access to ResearchAI
Jul 2026	Midline survey wave 1 (Month 3); mechanism variables collected as flows from baseline
Oct 2026	Endline survey wave 2 (Month 6); primary outcome (REI change index) measured
Oct 2026	Short-run ITT analysis; preparation of Stage 2 submission
Apr 2027	Waitlist control group receives access (crossover)

1 Introduction

A half-century of technological change has polarised labour markets by hollowing out middle-skill occupations while leaving the highest-skill positions intact, their returns sustained by the scarcity of the expert judgment they require ([Autor and Dorn, 2013](#); [Acemoglu and Restrepo, 2018](#)). AI sits at a historical inflection point. It could dissolve that scarcity and extend expert capability to a broader workforce, or it could further concentrate returns among those already at the frontier. Which path prevails is an empirical question, and it turns on whether AI can transmit expert judgment to those who lack it, or whether it merely amplifies the output of those who already possess it. This paper provides the first experimental evidence on that question in a prototypical high-expertise cognitive occupation, studying the causal effect of AI access on researcher productivity.

The central question of this paper is whether generative artificial intelligence can transmit expert judgment to researchers who do not yet possess it. That question is distinct from the one that has occupied the automation literature for a decade — whether AI substitutes for routine tasks — because it concerns not what the technology removes from the production process but what it adds to those who are furthest from the knowledge frontier. Answering it tells us whether AI equalizes or amplifies the human-capital inequality embedded in the structure of the profession.

[Autor \(2024\)](#) and [Autor and Thompson \(2025\)](#) provide the framework that identifies the central mechanism. The standard task-based model predicts that workers sharing the same automation exposure are similarly affected ([Acemoglu and Restrepo, 2018](#)). The expertise framework overturns this prediction: what determines the consequences of automation is not which tasks are removed, but whether the tasks that remain become more or less expert in their demands, and crucially, whether the expertise needed to perform them can be acquired or remains scarce. Generative AI introduces a complication that prior waves did not. It is capable of encoding judgment, not merely codifiable procedure, making it the first general-purpose technology that could in principle transmit expertise rather than merely substitute for it. Whether it does so—whether expert AI guidance diffuses the capacity for high-quality work or merely accelerates the production of those who already command it—is an empirical question that theory alone cannot resolve.

We study this question in academic economics, a setting in which expert knowledge is unusually concentrated and the unit of expert output is unusually well-defined. Access to the tacit standards that govern publication and intellectual influence in the discipline is slow to diffuse and systematically unequal across institutions and career stages—the hidden curriculum of the profession ([Heckman and Moktan, 2020](#)). Peer review sits at the heart of this concentration: the mechanism through which expert judgment is exercised and reproduced, and among the most cognitively demanding tasks in any knowledge economy. We have built and deployed ResearchAI, a tool that provides what general AI provides and adds the expert diagnosis that general AI delivers only to those who already possess the expertise to elicit it. A junior researcher using a general AI tool receives answers calibrated to the questions she knows to ask; a researcher using ResearchAI receives, in addition, a diagnosis of what she should be asking—where her identification strategy is vulnerable, whether her contribution is genuinely novel, whether her effects are economically significant—and concrete guidance on how to address it. The tool does not execute the research. It attempts to transmit directly to junior researchers the tacit standards that constitute the hidden curriculum, and

that general AI leaves locked behind the expertise frontier.

We test the expertise framework with a pre-registered randomised controlled trial in which junior economists are randomly assigned to use ResearchAI while advancing their own research. Both treated and control researchers retain access to general AI tools; only treated researchers receive the expert-intensive guidance of ResearchAI, so random assignment generates exogenous variation in the receipt of expert AI feedback at the margin where it is scarcest (Athey and Imbens, 2017). Our primary outcome is researcher productivity. Following the productivity tradition, we define productivity not as the quantity of output but as the *quality of research relative to the effort it requires*: a researcher is more productive when it becomes less effortful for her to produce work that meets a higher quality bar. We anchor the notion of quality in the criteria by which strong empirical economics papers are conventionally evaluated, under which a paper must be *credible* (the necessary condition: the question is answered with a state-of-the-art identification strategy) and must clear the sufficient condition of *originality* (it answers its research question better than the prior literature—the contribution) and *significance* (answering it helps solve a higher-order economic problem—the value added). We measure productivity through a set of individual survey items, each capturing the within-person change since baseline in how effortful it is to reach quality along one of these dimensions, plus an overall quality-for-given-effort item. The measures are deliberately subjective: objective output counts are uninformative for long-cycle knowledge production in a six-month window, and the perceived ease of producing high-quality work is the proximal margin on which an expert-feedback intervention is expected to act. We treat this within-person change in productivity as the proximal outcome of the intervention, the appropriate measure when the downstream output, a published paper, takes years to materialise. The exact measures are defined in Section 3.5.

This paper makes two contributions. The first is empirical. We provide experimental evidence on whether expert diagnostic guidance—AI whose primary function is to diagnose rather than to perform the work—diffuses or concentrates the returns to expertise within

a high-expertise profession. The automation literature has asked which tasks AI removes, and a growing body of field experiments has documented average productivity gains from general-purpose AI (Brynjolfsson et al., 2025; Noy and Zhang, 2023; Dell’Acqua et al., 2026), with gains that often accrue disproportionately to lower-skill workers. Because in each setting the AI executes the task, these designs cannot separate the two regimes the expertise framework distinguishes: transmission of expertise versus acceleration of those who already hold it. By randomizing access to expert-style diagnostic feedback at the margin where it is scarcest, our heterogeneity analysis identifies which regime holds. The second contribution is methodological. The democratization reading of our heterogeneity result — that ResearchAI transmits genuine expertise rather than merely substituting for the researcher’s own judgment — requires evidence that gains among less-expert researchers coincide with retained intellectual ownership of the resulting work. We measure this directly, asking treated researchers whether the contribution remains genuinely their own rather than the tool’s. This is, to our knowledge, the first randomized measurement of AI-induced intellectual ownership in a field experiment, and it speaks directly to concerns about generative AI raised in creative and academic settings but not previously subjected to experimental scrutiny.

This document is our Stage 1 Registered Report, submitted under the JPE Micro Registered Reports track. The trial is pre-registered on the AEA RCT Registry (AEARCTR-0017749), and the analysis plan set out in this document constitutes the binding commitment governing our endline analysis upon in-principle acceptance. Section 2 presents the ResearchAI platform and develops the conceptual framework and predicted mechanisms. Section 3 sets out the experimental design: treatments, randomization, sample, hypotheses, outcome variables, and statistical power. Section 4 describes the data sources, survey instruments, and timing, and reports balance and preliminary evidence. Section 5 sets out the empirical strategy: estimating equations, inference, multiple-testing correction, heterogeneity, and robustness. Section 6 discusses limitations, threats to validity, and implementation risks.

2 Context and Conceptual Framework

2.1 ResearchAI Development Suite

ResearchAI is expert diagnosis. A junior researcher using a general-purpose AI tool receives answers calibrated to the questions she knows how to ask. A researcher using ResearchAI receives, in addition, a diagnosis of what she should be asking — where her identification strategy is vulnerable, whether her contribution is genuinely novel relative to the frontier, whether her effects are economically significant — and a structured roadmap for how to address it. The tool delivers expert judgment on precisely the dimensions that determine whether a paper is published or rejected: credibility, originality, and significance. This is the empirical anchor for the transmission channel we study. General-purpose AI substitutes for the codifiable, context-independent components of research; ResearchAI attempts to transmit the evaluative standards that constitute the hidden curriculum of the profession (Heckman and Moktan, 2020), standards that early-career researchers have not yet internalised and that general AI delivers only to those who already possess the expertise to elicit them. Such expert diagnosis is not freely available today. It can be obtained only by paying for specialist research-feedback services or by having the skill to extract it from general-purpose tools through careful prompting — which is precisely the skill that defines the expertise gap we study.

The evaluative standards encoded in ResearchAI were derived from a systematic comparison of working papers with their subsequently published versions in top-tier economics journals (AER, JPE, Econometrica, QJE, ReStud, and leading field journals). The delta between the pre-publication draft and the final accepted version reveals the expert judgment that the refereeing process demanded: the identification threats the authors had not recognised, the contribution they had not articulated, the economic significance they had not established. This distance — the gap between what the researcher produced unaided and what the editorial process required — encodes the tacit knowledge that the hidden curricu-

lum transmits slowly and unequally across institutions and career stages. Three systematic patterns emerged from this calibration exercise. First, working papers frequently contained identification threats that the authors did not recognise; the published versions had resolved them, but only after multiple rounds of revision guided by expert referees. Second, drafts routinely failed to articulate why their findings constituted an advance over the existing literature; the published versions had crystallised the contribution. Third, working papers answered technical questions without connecting them to higher-order economic problems; the published versions had established that connection. The common bottleneck across all three dimensions was not researcher effort but access to the evaluative judgment that would diagnose what was binding — precisely the resource that is scarce for researchers distant from the knowledge frontier. ResearchAI codifies these revealed standards of the editorial process and delivers them directly to the researcher’s own manuscript.

The platform implements a multi-agent orchestration architecture that decomposes the complex, non-routine cognitive task of manuscript evaluation into discrete audits handled by specialised agent personas. Each agent evaluates a specific dimension — identification credibility, contribution novelty, economic significance, expositional clarity — and their outputs are synthesised into a single prioritised diagnostic report. The system works on the researcher’s own manuscript, providing iterative feedback calibrated to the standards extracted from the working-paper-to-publication comparison described above.

Three design features are calibrated to force human capital accumulation and prevent the passive delegation documented in the automation literature. First, the synthesised roadmap categorises feedback into fatal econometric flaws, major revisions, and minor prose polish, visually prioritising threats to internal validity over typographical corrections. This mitigates the bandwidth constraints of junior researchers and directs attention toward high-return analytical tasks. Second, the system embeds anti-delegation frictions: if a researcher requests cosmetic rewrites while a fatal identification flaw remains unresolved, the tool refuses prose optimisation and redirects attention to the core econometric issue, ensuring it

operates as a pedagogical complement rather than a substitute for the researcher’s own judgment. Third, the platform tracks manuscripts across multiple uploads and generates version comparisons highlighting resolved versus persistent issues. The grading rubric is calibrated to the rejection standards of top-tier journals, explicitly capping scores for papers with unresolved identification threats, which prevents the sycophancy characteristic of general-purpose LLMs and provides researchers with an uncontaminated signal of manuscript quality.

Platform usage is rivalrous by design. Treatment exposure is constrained through *Active Project Slots*, which cap the number of simultaneous manuscripts a user can process: two for doctoral candidates, mirroring the binding constraints of the modern three-article dissertation, and three for junior faculty. The cap directs the tool toward the user’s most critical, job-market-relevant outputs. The capacity constraint imposes a prohibitive opportunity cost on account sharing: a treated student who processes a control-group peer’s manuscript must permanently delete the iterative history of one of her own chapters, a trade-off the convexity of job-market payoffs renders unattractive.

2.2 How ResearchAI transmits expert knowledge: the causal pathway

Figure 1 maps the mechanism through which ResearchAI is hypothesised to raise research productivity. The pathway has two channels, both originating from the same intervention but operating on different margins of the production function.

The *direct channel* is the transmission of expert evaluative standards. The multi-agent diagnosis identifies what is binding in the manuscript — identification threats the researcher has not recognised, a contribution that has not been articulated, economic significance that has not been established — and delivers this judgment without requiring the researcher to know what to ask. This is the channel through which tacit knowledge, ordinarily acquired through years of exposure to senior colleagues, referee reports, and editorial decisions, is transmitted directly.

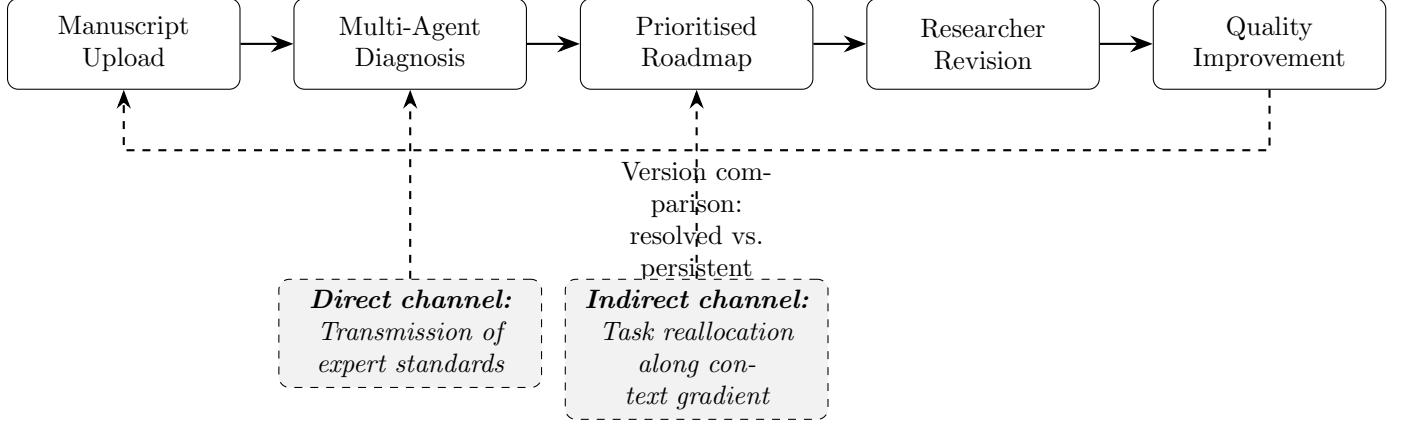


Figure 1: Causal pathway from ResearchAI access to productivity improvement. The direct channel transmits the evaluative standards of the knowledge frontier through diagnostic feedback the researcher did not know to request. The indirect channel reallocates researcher effort from low-context, codifiable tasks toward the high-context, judgment-intensive work that determines publication quality. The feedback loop — re-upload, version comparison, updated diagnosis — is the iterative mechanism through which standards are internalised rather than passively consumed.

The *indirect channel* is task reallocation along the context gradient. By taking over the low-context components of manuscript evaluation (checking econometric specification against known threats, surveying related literature, auditing expositional clarity) and delivering a prioritised roadmap, the system lowers the effort cost of the high-context, judgment-intensive work that determines a paper’s quality: refining the conceptual framing, evaluating whether the identification strategy is credible, judging whether the question is genuinely novel and significant. The productivity gain we study therefore runs through making the high-value work easier — not through time freed up by automating the routine, since both treatment and control groups retain access to general-purpose AI for that purpose.

The feedback loop — re-uploading revised manuscripts, receiving updated diagnoses, comparing resolved and persistent issues across versions — is the iterative mechanism through which the transmitted standards are internalised rather than passively consumed. The anti-delegation frictions described above ensure that this loop operates as a pedagogical complement: the researcher must engage with the substance of the diagnosis to make progress, and the system refuses to substitute for the analytical work that remains.

2.3 Conceptual framework and predicted mechanisms

The central theoretical question concerns which of the two channels described above dominates the productivity effect, and for whom. Following the expertise framework of [Autor and Thompson \(2025\)](#), we characterise research tasks by their *context content*: the degree to which performing them requires domain-specific knowledge that cannot be encoded and applied generically. Tasks low in context content — collecting and preparing data, writing and debugging code, executing analyses, reading and synthesising papers — are the tasks for which AI substitutes most readily. Tasks high in context content — generating ideas, developing and refining the conceptual framing, evaluating identification credibility, and judging whether a question is genuinely novel and significant — are complementary to expertise and resist substitution; these are precisely the activities that determine a paper’s credibility, originality, and significance. The prediction we test is therefore a *relative shift*: treated researchers should report doing the high-context, judgment-intensive tasks more frequently relative to the low-context ones. The productivity gain runs through making the high-value work easier — not through time freed up by automating the routine.

This generates two opposite predictions about which researchers benefit most. In the *complement* case, converting the diagnosis into higher-quality output requires a stock of accumulated expert knowledge, so the gains are largest for researchers already closer to the knowledge frontier; AI amplifies existing expertise. In the *substitute* case, the tool itself transmits frontier standards to those who lack them — through the direct channel of repeated diagnostic feedback calibrated to top-journal norms — so the gains are largest for the least expert; AI democratises expert judgment. This dichotomy is the micro, within-person counterpart of the macro mechanism in [Autor and Thompson \(2025\)](#), where automation that lowers the expertise required for the remaining tasks lets less-expert workers enter, while automation that raises it concentrates returns among incumbents; their object of study is wages and employment across occupations over four decades, whereas ours is the change in an individual researcher’s productivity, but the underlying force — whether removing tasks raises or

lowers the expertise the remaining work demands — is the same. Our heterogeneity analysis, set out in the experimental design (Section 3), is exactly a test of whether the system complements or substitutes for a researcher’s pre-existing expertise, with direct implications for whether AI equalises or amplifies existing human-capital inequality.

3 Experimental Design

3.1 Treatment Rollout and Step-Wedge Design

The experiment employs a waitlist-control design. Following randomization at $t = 0$, the Treatment group receives immediate access to the ResearchAI suite. The Waitlist Control group conducts research as usual — which may include ad-hoc use of general-purpose commercial LLMs — before crossing over to receive full access to the suite after the 6-month baseline survey is collected. This guarantees all participants eventual access while preserving a clean treatment-control contrast over the 0–6 month window during which our primary outcome is measured.

To minimize differential attrition, the waitlist guarantee is explicitly communicated to the control group prior to randomization, and participants in the control arm receive single-use tokens upon completing verified survey waves at months 3 and 6. A token grants access for the upload and review of a single document, providing a minimal, discrete benefit without conferring the continuous, iterative access that characterizes the treatment condition. This incentive structure maintains engagement in the control arm while preserving a clean contrast between the treatment and control conditions during the 0–6 month window of clean identification.

The design provides a clean Intention-to-Treat (ITT) contrast over the 0–6 month window for our primary outcome (research productivity) and our mechanism outcomes. To address non-compliance — treated participants who do not engage with the tool — we complement the ITT estimates with a Local Average Treatment Effect (LATE) for active users,

instrumenting platform usage with random assignment, as described in Section 5

3.2 Randomization

Participants were randomly assigned to the Treatment arm or the Waitlist Control arm in a 1:1 ratio.

3.3 Sample selection and eligibility

The target population comprises early-career economists: current PhD students and junior faculty (assistant professors and postdoctoral researchers) within five years of their doctorate. The restriction to early-career researchers serves both substantive and methodological purposes. Substantively, the tacit knowledge problem that motivates our study binds most tightly at this career stage: the norms and heuristics of top-tier publication are incompletely internalized and still actively being acquired.

Interested individuals completed a screening questionnaire to verify eligibility, followed immediately by a comprehensive baseline survey capturing demographic characteristics, historical productivity metrics, baseline AI usage patterns, and subjective well-being indicators. Because participants recruited early in the window may differ systematically from late adopters in baseline effectiveness or technological familiarity, we record the exact date of enrollment to test for differential selection.

Building on the framework developed in Section 2, we formalize a set of hypotheses linking treatment assignment to the change in our primary outcome and to the mechanism variables. Hypotheses 1 and 2 constitute the confirmatory core of the pre-analysis plan.

3.4 Hypotheses

Hypothesis 1 (Average Treatment Effect on Productivity): *Access to structured, multi-agent AI feedback significantly increases researcher productivity, as measured by a pre-*

specified family of individual survey items that capture how effortful it is to produce research, relative to the waitlist control group.

We test H1 by estimating the average treatment effect β in Equation (3) below separately for each productivity item. Our headline confirmatory endpoint is the overall *quality-for-given-effort* item (the effort required to bring the main paper up to the quality standards of the target journal).

Hypothesis 2 concerns the mechanism developed in Section 2: the reallocation of research effort along the context gradient.

Hypothesis 2 (Mechanism: Time Substitution Along the Context Gradient).

Treatment reallocates research effort up the context gradient. Treated researchers report a larger increase in the frequency of high-context, judgment-intensive tasks than in the frequency of low-context, codifiable tasks. The key test is the relative shift: the differential movement of effort from the bottom to the top of the context gradient, which mediates the effect of treatment on productivity.

We locate each researcher on the knowledge gradient using three pre-treatment proxies available in our baseline data, used jointly. *Career stage* — PhD student versus junior faculty or postdoctoral researcher — captures cumulative exposure to the evaluative standards of the frontier. *Baseline mentoring access* captures the quality of the researcher’s counterfactual escalation channel, namely how much expert feedback she receives absent the tool. *Institutional tier* captures the density of the expert environment she is embedded in. The researcher’s position on the gradient allows us to distinguish between the democratisation and skill-complementing predictions without relying on any single imperfect measure.

Formally, the interaction specification underlying is:

$$\Delta Y_i = \alpha + \beta_1 T_i + \beta_2 M_i + \beta_3 (T_i \times M_i) + X_i' \delta + \varepsilon_i \quad (1)$$

where T_i is the treatment indicator, M_i is the pre-specified expertise moderator, ΔY_i is the

change in productivity or in the relevant mechanism measure from baseline, and X_i is a vector of baseline covariates including the stratification fixed effect. Because self-assessed expertise is endogenous to the treatment, M_i is built from the three pre-treatment proxies described above — career stage, baseline mentoring access, and institutional tier — each coded to increase in expertise and used jointly, either as a standardized composite or by entering the three interactions together. Under this orientation, a *negative* β_3 is consistent with the skill-leveling / substitute channel (H2a), since the gains then fall as baseline expertise rises; a *positive* β_3 is consistent with the skill-complementing / complement channel (H2b). All interaction tests are pre-specified but exploratory.

A guardrail accompanies the heterogeneity test. The democratisation reading of H2a requires that the tool transmit genuine expertise rather than merely substitute for the researcher’s own judgment. We therefore measure *intellectual ownership* — whether the researcher retains the sense that the resulting contribution is genuinely her own rather than the tool’s — as a check that any gains reflect skill formation rather than outsourcing. This is a concern raised about generative AI in creative and academic settings but, to our knowledge, not yet measured in a randomized field experiment. It is reported as a secondary guardrail, not a confirmatory mechanism.

3.5 Outcome variables

The classification of tasks into high- and low-context indices follows strictly the criterion of [Autor and Thompson \(2025\)](#): a task has high context content if executing it well requires domain-specific expert knowledge that cannot be codified and applied generically; it has low context content if the instructions for executing it can be made sufficiently explicit that someone without accumulated disciplinary expertise can perform it correctly. Tasks that are theoretically ambiguous under this criterion — because their context content depends on the specific content of the activity rather than the activity type — are excluded from both confirmatory indices and reported individually as standalone items. This exclusion is

pre-specified and does not affect the headline test of H1, which uses the productivity items (PQ1–PQ5) directly.

Research productivity items — primary outcome. We measure productivity as the quality of a researcher’s work relative to the effort it takes to reach it: a researcher is more productive when she can clear a higher quality bar with less effort. Five items (Table A.1) capture this, each asking how things have changed since the start of the study. Four ask whether it has become *easier* to reach quality (PQ1-PQ4). A fifth item, PQ5, asks the converse, for the *same* amount of work, whether the overall quality the researcher can produce has risen. Each item is measured at the 3-month and 6-month waves as a flow relative to baseline.

High-context task frequency — mechanism. Change since baseline in the frequency of high-context, judgment-intensive activities whose performance requires accumulated expert knowledge. These are five items (E1–E5, Table A.2): generating research ideas or hypotheses (E1); developing or refining the conceptual framing or contribution of a project (E2); discussing research progress with coauthors (E3); leading the overall direction of a project (E4); and planning or coordinating tasks and timelines (E5). Each item is identical in content to the corresponding baseline task-frequency item and is measured as a flow. Under H2, treatment raises the frequency of these tasks relative to the low-context tasks below.

Low-context task frequency — mechanism. Change since baseline in the frequency of low-context, codifiable activities for which AI substitutes most readily. **These are five items (R1–R5, Table A.2):** collecting, cleaning, or preparing data (R1); writing, debugging, or improving code and running analyses (R2); writing or revising research papers or drafts (R3); editing or improving the clarity, structure, and presentation of written work (R4); and reading and synthesising academic papers (R5). Following Autor and Thompson (2025), the prediction is a reallocation of effort up the context gradient; the test is therefore the *relative* movement of the high-context versus low-context tasks. We make no signed prediction for either set of tasks in isolation, but predict that the high-context minus low-context difference

risers under treatment.

Intellectual ownership. Three items capturing the researcher’s sense that the resulting work is genuinely her own contribution rather than the tool’s: ownership of ideas, ownership of writing, and willingness to claim authorship without qualification. Reported item by item, this guardrail discriminates between the two readings of the heterogeneity result: democratisation (H2a) is credible only if gains among less-expert researchers coincide with retained ownership, indicating genuine skill formation rather than outsourced judgment.

Objective proxies — secondary outcomes. Self-reported submission attempts (item P1) and new drafts or projects initiated (item P2), both collected at the 6-month endline as counts since the start of the study, and ordinal project progress (item P3), collected at the 3- and 6-month waves, together with platform usage intensity and revision cycles from administrative logs (the *Usage intensity* and *Iterations per manuscript* measures in Table A.2, available for the treatment arm). The exact wording of items P1–P3 is given in the survey instruments (Table A.1). These complement the perception-based primary outcomes with behavioral measures.

The full mapping from hypotheses to statistical tests, including the MDES for each test, is provided in Section 3.6 (Power Calculations) and summarized in Table 2. Exact item wording for all measures, together with the full midline (3-month) and endline (6-month) survey instruments, is provided in Table A.1.

3.6 Power Calculations

Our primary confirmatory hypothesis (H1) requires estimating the average treatment effect β in Equation 3 for each research-productivity item, with the overall quality-for-given-effort item (PQ4) as the headline endpoint. The minimum detectable effect size (MDES), at 80 percent power and $\alpha = 0.05$ (two-sided), follows:

$$\text{MDES} \approx (t_{1-\kappa} + t_{\alpha/2}) \times \left[\frac{1}{N \times P(1-P)} \right]^{1/2} \times \sigma \times \sqrt{1-R^2} \quad (2)$$

where $N = 148$ (75 treatment, 73 control), $P \approx 0.51$, $\sigma = 1$ (standard deviation of the standardized outcome), and $(t_{0.8} + t_{0.025}) \approx 2.80$. Without covariates, the MDES is $2.80/\sqrt{148 \times 0.51 \times 0.49} \approx 0.46$ standard deviations. Each productivity item is a within-person flow, and the included baseline covariates—career stage, field, baseline task frequencies, baseline AI use and confidence—absorb a substantial share of outcome variance. Under a conservative $R^2 = 0.30$ the effective MDES falls to approximately 0.39 standard deviations; exploiting the full three-wave panel (the 3-month and 6-month flows) with individual fixed effects ($R^2 \approx 0.50$) reduces it further to approximately 0.33 standard deviations. These values lie within the range of effect sizes documented for AI interventions on perceived competence and work quality, which range from 0.40 to 0.65 standard deviations (Brynjolfsson et al., 2025; Noy and Zhang, 2023; Dell’Acqua et al., 2026). The study is therefore adequately powered for its primary confirmatory test under both the panel and cross-sectional specifications. Because the primary productivity items are tested as a family, we report FDR-adjusted q -values (Anderson, 2008) alongside the per-item p -values; the headline endpoint (PQ4) is tested without adjustment.

The same formula applies to our mechanism outcomes (the high-context and low-context task-frequency items), which share the lagged-dependent-variable structure—each has an identical baseline level—and thus achieve comparable MDES values of approximately 0.39 standard deviations; the key relative (high-context minus low-context) contrast is estimated with comparable precision.

By contrast, the heterogeneity interactions in Equation (1) are constrained by the size of the smallest cell. With a roughly even split on each binary moderator, each treatment-by-moderator cell contains approximately $N = 37$, yielding an MDES of approximately 0.78 standard deviations. The central tests of Hypotheses 2a and 2b—whether the treatment-by-expertise interaction is negative (skill-leveling) or positive (skill-complementing)—are there-

fore powered to detect only large effects. We pre-specify these tests as exploratory for this reason: a null result at this sample size cannot distinguish between the absence of heterogeneity and insufficient power to detect a moderate effect, and we will interpret the interaction estimates accordingly. The results are intended to discipline the direction of future, adequately powered work rather than to settle the distributional question. Table 2 summarizes MDES estimates across all outcomes and specifications.

Table 2: Minimum Detectable Effects by Hypothesis and Specification

Hypothesis	Estimand	N per arm	Total N	MDES (SDs)
H1: ATE on productivity (headline PQ4)	β , Eq. (3)	75/73	148	0.39
H1: ATE on productivity (3-wave panel, FE)	β , Eq. (3)	75/73	148	0.33
H2: ATE on high-context task frequency	β , Eq. (3)	75/73	148	0.39
H2: ATE on low-context task frequency	β , Eq. (3)	75/73	148	0.39
H2: ATE on high-context – low-context (key)	β , Eq. (3)	75/73	148	0.39
H2a/2b: T $\times$ Expertise composite (key)	β_3 , Eq. (1)	~ 37 /cell	148	0.78
H2a/2b: T $\times$ Career stage	β_3 , Eq. (1)	~ 37 /cell	148	0.78
H2a/2b: T $\times$ Mentoring access	β_3 , Eq. (1)	~ 37 /cell	148	0.78
H2a/2b: T $\times$ Institutional tier	β_3 , Eq. (1)	~ 37 /cell	148	0.78

Notes: MDES computed at 80% power, $\alpha = 0.05$ (two-sided), $\sigma = 1$. Average-effect rows assume baseline covariate adjustment with $R^2 = 0.30$ (cross-sectional) or $R^2 = 0.50$ (three-wave panel with individual fixed effects). Interaction rows use the smallest treatment-by-moderator cell size (~ 37), assuming an even split on each binary moderator; these tests are powered only for large effects and are pre-specified as exploratory. The treatment-by-expertise interaction—with expertise as the standardized composite of career stage, mentoring access, and institutional tier—is the key test of skill-leveling versus skill-complementing (H2a/H2b); the three single-proxy interactions are its components. All MDES values are in standard deviations of the standardized outcome.

Power under observed compliance. The MDES values above assume full take-up. Platform logs from the initial rollout indicate an observed compliance rate of approximately $\pi \approx 0.70$, with around 30 percent of treated participants not yet engaging above a minimal threshold of one completed feedback cycle. This has two implications for the power of the study, both of which we account for here.

First, it affects the ITT estimate. The ITT averages over compliers and non-compliers and is therefore attenuated relative to the effect on active users. With a compliance rate of $\pi \approx 0.70$, the relationship between the ITT and the Local Average Treatment Effect (LATE) is $\text{LATE} = \text{ITT}/\pi$, so a LATE of 0.39 SD would appear as an ITT of approximately 0.27 SD.

The ITT remains the primary policy-relevant estimand — it measures the effect of *offering* access — but we report both.

Second, it affects the effective power of the ITT test. The observed compliance rate implies that the ITT MDES computed in Section 3.6 — which assumed full take-up — is conservative; the true MDES for the ITT is approximately $0.39/0.70 \approx 0.55$ SD in the cross-sectional specification with $R^2 = 0.30$. Table 4 reports MDES values under the observed compliance rate for the headline ITT and LATE.

4 Data

4.1 Data sources, instruments, timing

The analysis draws on three complementary data sources, linked at the participant level. First, survey data collected at three waves: a baseline survey administered prior to randomization, capturing demographic characteristics, historical productivity, baseline AI usage, tasks frequency and subjective well-being, and follow-up surveys at month 3 (midline) and month 6 (endline), which capture the primary outcome and the mechanism variables, with the 6-month wave serving as our endline. Second, platform log data generated by the ResearchAI server and not accessible to participants, recording engagement at the manuscript level: the number of uploads per month, the frequency of agent interactions per paper, the cumulative number of iterative uploads and version comparisons, and the number of active project slots used. Third, administrative data: bibliographic records matched from ORCID and Google Scholar with participant consent, used to construct objective output measures such as working-paper circulation. The full instrument across waves is reproduced in Table A.1, and the construction of every analysis variable is detailed in Table A.2.

4.2 Balance checks and preliminary evidence

We assess covariate balance using the baseline data available to date. Table 3 reports baseline summary statistics and covariate balance by treatment status, in the subsample for which baseline data are currently available. The sample is broadly balanced across treatment and control groups. Two variables — specialization in macroeconomics and use of AI for writing — show nominally significant differences between arms; however, the joint F -test of all covariates on treatment assignment, controlling for stratification fixed effects, fails to reject the null of overall balance. All specifications will include stratification fixed effects and the relevant baseline covariate as controls, ensuring these differences do not confound treatment effect estimate.

5 Empirical Strategy

5.1 Statistical model and estimating equations

Our primary specification estimates the average causal effect of access to ResearchAI on the change in productivity from baseline. Because all outcomes are measured as flows (changes from baseline), our preferred estimator is:

$$\Delta Y_{i,t} = \alpha + \beta \text{Treatment}_i + X_i' \delta + \varepsilon_{i,t} \quad (3)$$

where $\Delta Y_{i,t} = Y_{i,t} - Y_{i,\text{baseline}}$ is the change in the outcome of interest from baseline to survey wave t (months three and six), Treatment_i is a binary indicator for treatment assignment, and X_i' is a vector of baseline covariates including the stratification fixed effect (career stage). Standard errors are heteroskedasticity-robust. The coefficient β identifies the ITT effect of being assigned to receive immediate access to ResearchAI on the change in the outcome relative to baseline.

Table 3: Baseline Balance Table

Variable	(1) Control	(2) Treatment	(3) Difference	(4) P-value	(5) N
Panel A: Demographics and Research Fields					
Female (=1)	0.521	0.507	-0.015	0.858	148
Full time student/researcher (=1)	0.973	0.947	-0.026	0.426	148
Native English speaker (=1)	0.096	0.173	0.078	0.169	148
Has caring responsibilities (=1)	0.164	0.160	-0.005	0.934	148
Parents completed a university degree (=1)	0.592	0.653	0.062	0.443	146
Field: Microeconomics (=1)	0.329	0.267	-0.061	0.410	148
Field: Macroeconomics (=1)	0.096	0.200	0.104*	0.075	148
Field: Econometrics (=1)	0.123	0.093	-0.030	0.565	148
Field: Labour/Public (=1)	0.452	0.453	0.002	0.982	148
Field: IO/Finance/Trade (=1)	0.110	0.080	-0.030	0.545	148
Field: Political Economy (=1)	0.082	0.147	0.065	0.221	148
Field: Other (=1)	0.274	0.213	-0.061	0.394	148
Career mentoring access (1 = strongly disagree, 5 = strongly agree)	3.425	3.627	0.203	0.199	148
Panel B: Academic and Publication Profile					
Total peer-reviewed papers	1.589	1.733	0.140	0.226	148
Papers awaiting decision	0.795	0.893	0.093	0.616	148
Papers currently working on for submission	2.041	2.413	0.370	0.168	148
Papers published in Top 7 journals	0.151	0.013	-0.137	0.275	148
Papers to submit next month	0.822	0.853	0.030	0.828	148
Number of active research coauthors	4.041	3.907	-0.149	0.750	148
Active coauthors more junior than you	0.740	0.467	-0.278**	0.047	148
Active coauthors more senior than you	2.699	2.987	0.282	0.451	148
Satisfaction with work-life balance (1 = very dissatisfied, 5 = very satisfied)	3.247	3.453	0.207	0.228	148
Panel C: Baseline AI Tools Usage					
Uses AI in research workflow (=1)	0.986	0.973	-0.013	0.573	148
Type of AI tools used	1.068	1.187	0.118	0.126	148
Freq. of 2-3 hr blocks (1-5 scale)	4.096	4.000	-0.093	0.527	148
Confidence using AI tools effectively (0-10)	6.644	6.293	-0.348	0.260	148
AI Use: Ideation and feedback (=1)	0.534	0.533	-0.000	0.998	148
AI Use: Writing (=1)	0.890	0.827	-0.064	0.264	148
AI Use: Background research (=1)	0.507	0.453	-0.053	0.522	148
AI Use: Data analysis (=1)	0.493	0.560	0.068	0.407	148
AI Use: Data analysis interpretation (=1)	0.315	0.280	-0.034	0.648	148
AI Use: Coding (=1)	0.877	0.840	-0.036	0.525	148
AI Use: Mathematical derivations (=1)	0.205	0.227	0.021	0.755	148
AI Use: I don't use AI (=1)	0.014	0.027	0.013	0.573	148
Panel D: Frequency of Research Tasks (1 = never, 5 = very often)					
Collecting, cleaning, or preparing data	3.425	3.560	0.138	0.466	148
Writing, debugging code, running analyses	3.808	3.813	0.007	0.970	148
Writing or revising papers or drafts	3.836	3.587	-0.248	0.194	148
Editing or improving clarity of written work	3.849	3.587	-0.263	0.173	148
Generating research ideas/hypotheses	2.863	2.787	-0.073	0.720	148
Reading/synthesizing academic papers	3.192	2.987	-0.204	0.246	148
Developing/refining conceptual framing	3.288	2.987	-0.300	0.123	148
Discussing research with coauthors	3.205	3.427	0.220	0.272	148
Planning/coordinating tasks/timelines	3.178	3.160	-0.018	0.928	148
Leading funding application/proposal	1.781	1.760	-0.022	0.913	148
Leading overall direction of a project	2.904	3.013	0.106	0.661	148
Joint F-test (p-value)[†]	0.000***				

Notes: Columns (1) and (2) report raw sample means. Column (3) reports the coefficient on the treatment indicator from an OLS regression of the covariate on treatment, controlling for stratification fixed effects with heteroskedasticity-robust standard errors; column (4) reports the associated p-value. [†] The joint F-test p-value is from an omnibus OLS regression of the treatment indicator on all covariates simultaneously, controlling for stratification fixed effects. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The following variables collected at baseline are not included because they are not amenable to balance testing: participant name and surname; institutional e-mail address and ORCID identifier; Google Scholar profile link; PhD start year; names of active coauthors; and current institution.

5.2 Heterogeneous effects

We test the time-substitution mechanism of Hypothesis 2 with two complementary specifications. The first tests the average mechanism effect in the full sample by estimating Equation (3) with the high-context and low-context task-frequency measures as outcomes. The evidence for the context-gradient channel is the *relative* shift: a treatment effect that is more positive for the high-context tasks than for the low-context tasks, equivalently a positive effect on the high-context minus low-context difference, indicating that effort moves up the context gradient. We make no signed prediction for either set of tasks taken alone.

The second specification tests heterogeneity in the treatment effect, which is our key test of skill-leveling versus skill-complementing. We estimate:

$$Y_{it} = \alpha + \beta_1 \text{Treatment}_i + \beta_2 M_i + \beta_3 (\text{Treatment}_i \times M_i) + X_i' \delta + \varepsilon_{it} \quad (4)$$

where M_i is the pre-specified expertise moderator and Y_{it} is, in our central specification, the change in productivity (the headline quality-for-given-effort item; we also report the interaction for each of the remaining productivity items). Because self-assessed expertise is endogenous to the treatment, we do not measure M_i from post-treatment or ability-perception items. Instead we locate each researcher on the knowledge gradient using three pre-treatment proxies, all collected in the baseline survey and used jointly: career stage, from the baseline academic-position item (PhD student versus junior faculty or postdoctoral researcher), baseline mentoring access, from the baseline agreement item on access to people who help the researcher navigate research and academic careers (the 1–5 scale reported in Table 3); and institutional tier, derived from the institution the participant reports at baseline; each coded to increase in expertise. Our preferred construction enters the three proxies as a single standardized composite Expertise_i ; we also report the specification that enters the three interactions together. A negative β_3 supports the skill-leveling / substitute channel (Hypothesis 2a): researchers lowest on the gradient gain most because the tool transmits

the frontier standards they otherwise lack. A positive β_3 supports the skill-complementing / complement channel (Hypothesis 2b). We acknowledge that these tests are powered to detect only large effects. A null result will not, on its own, adjudicate between the channels, and we will interpret the joint pattern across the average mechanism effect (from Equation 3) and the interaction estimates accordingly.

5.3 Attrition and Non-Compliance

We distinguish two threats to the study’s statistical power that require separate treatment: survey attrition and treatment non-compliance.

Survey attrition. We anticipate overall attrition below 15 percent given the academic setting, the short six-month horizon, and the explicit commitment to full tool access for the control group upon study completion. Two design features limit differential attrition across repeated survey waves. First, the waitlist guarantee provides a credible, high-value commitment to all control-group participants, substantially reducing incentives to drop out before accessing the tool. Second, the single-use token incentive structure at each midline wave rewards continued engagement by the control arm without conferring the continuous, iterative access that characterizes the treatment.

We pre-specify the following decision rules, each evaluated independently. First, we will test for differential attrition by regressing an indicator for survey completion on the treatment indicator and all baseline covariates. If attrition exceeds 10 percent in either arm, we will report Lee (Lee, 2009) bounds alongside the main estimates. The Lee bounds require a monotonicity condition: that assignment to treatment weakly increases (or weakly decreases) the probability of survey completion in every stratum. We assume weak monotonicity in the positive direction — that treatment assignment does not reduce the completion probability for any participant type — on the grounds that immediate access to a valued tool raises, or at worst leaves unchanged, the incentive to remain in the study. This assumption would be violated if treatment-arm participants, having already received access, felt less compelled to

complete the survey; the waitlist-contingent token structure in the control arm works against this by making survey completion instrumentally valuable in both arms. We will report the completion rates by arm and stratum so that the reader can assess the plausibility of this assumption directly. Second, if the joint F -test of baseline covariates on completion rejects at $\alpha = 0.10$, we will additionally report inverse-probability-weighted (IPW) estimates using predicted completion probabilities from the first-stage attrition regression, regardless of the overall attrition level. The two rules operate independently: it is possible that attrition is low overall but compositionally non-random (triggering IPW but not Lee bounds), or that attrition is high but balanced across arms (triggering Lee bounds but not IPW).

The randomized analysis sample comprises $N = 148$ participants (75 treatment, 73 control), all of whom completed the baseline survey and met the eligibility criteria prior to randomization. Balance is shown on this full sample in Table 3.

Treatment non-compliance. Non-compliance — treated participants who do not engage with the ResearchAI platform — is a distinct threat from attrition. Based on platform log data collected during the initial treatment rollout, approximately 30 percent of treated participants have not yet engaged with the tool above the minimal threshold of at least one completed feedback cycle. This non-compliance rate has two implications that we address explicitly.

We pre-specify the following strategy to address non-compliance. We will characterize compliers at baseline to test whether non-compliance is random or systematic. If compliers and non-compliers differ significantly on baseline covariates, we will report complier average characteristics following Angrist et al. (1996) and test whether the compliance rate itself varies by career stage or institutional tier, since systematic non-compliance by subgroup would confound the heterogeneity analysis. We will define compliance on both the extensive margin (at least one completed feedback cycle) and the intensive margin (cumulative uploads above the sample median among active users), and report the LATE under both definitions as specified in Section 5.

Table 4: MDES Under Observed Compliance Rate ($\pi = 0.70$)

Estimand	Compliance	N	Spec.	MDES (SDs)
ITT (headline PQ4)	100% (assumed)	148	$R^2 = 0.30$	0.39
ITT (headline PQ4)	70% (observed)	148	$R^2 = 0.30$	0.55
ITT (headline PQ4)	70% (observed)	148	Panel FE	0.47
LATE (compliers)	70% (observed)	148	$R^2 = 0.30$	0.55
LATE (compliers)	70% (observed)	148	Panel FE	0.47

Notes: All rows use the full analysis sample $N = 148$ (75 treatment, 73 control). ITT rows at 100% compliance report the benchmark MDES under full take-up. ITT rows at 70% compliance report the attenuated ITT MDES: because the ITT averages over compliers and non-compliers, the detectable effect on the treated population must be $1/\pi = 1/0.70$ times as large to register in the ITT, yielding $\text{MDES}_{\text{ITT}} = 0.39/0.70 \approx 0.55$. LATE rows report the MDES for the effect *on compliers* recoverable via 2SLS from the full sample. The Wald / 2SLS estimator divides the reduced-form ITT by the first-stage compliance rate π ; the MDES for the LATE therefore equals the ITT MDES, since both the numerator and denominator are scaled by the same π (Bloom, 1984). The LATE is estimated on all $N = 148$ observations, not on the complier subsample alone; power is governed by the first-stage F -statistic, which we report alongside each 2SLS estimate. $R^2 = 0.30$ denotes the cross-sectional specification with baseline covariate adjustment; Panel FE denotes the three-wave panel with individual fixed effects ($R^2 \approx 0.50$). All values at 80% power, $\alpha = 0.05$ (two-sided).

An additional consideration for the mechanism test (H2) is that non-compliance may be non-random with respect to the moderator. If non-compliers are disproportionately drawn from one career-stage stratum, the effective cell size for the treatment-by-expertise interaction will be smaller than the $N \approx 37$ reported in Table 2, further reducing power for H2a and H2b. We will characterize the compliance rate by stratum and report this as part of the pre-analysis.

5.4 Non-Compliance: Local Average Treatment Effect

The ITT estimate is the policy-relevant effect of *offering* access to ResearchAI, and constitutes our primary reported result. Because some treated participants may not actively engage with the tool, the ITT averages over compliers and non-compliers, potentially understating the effect for those who actually used the platform. We therefore complement the ITT with a Local Average Treatment Effect (LATE) for active users, using random as-

signment as an instrument for adoption. We capture engagement along two complementary margins and report the LATE under both, with a pre-specified primary measure for each.

Extensive margin (primary LATE). On the extensive margin, $\text{Active}_i \in \{0, 1\}$ is a dichotomous adoption indicator equal to one for participants who complete at least one full feedback cycle on the platform (uploading a manuscript, receiving the multi-agent diagnostic report, and engaging with the synthesized roadmap). This is our pre-specified primary definition of compliance; under it the estimand is the standard complier LATE / CACE.

Intensive margin: treatment intensity as cumulative engagement time. On the intensive margin, we pre-specify as the primary dose measure the cumulative number of completed feedback cycles over the study window, $\text{Dose}_i \in \mathbb{N}_{\geq 0}$, constructed from platform server logs. Each feedback cycle — uploading a manuscript, receiving the diagnostic report, and re-uploading a revised version — constitutes one unit of dose. This measure captures the depth of iterative engagement with the structured feedback, which is the proximal channel through which the tool is hypothesized to transmit expert judgment. We standardize Dose_i to have mean zero and unit standard deviation in the treatment arm, so that β in the second-stage equation is interpreted as the LATE per standard-deviation increase in engagement intensity among compliers. Because the model is just-identified by the single binary instrument, the intensive-margin LATE identifies a weighted average of marginal effects across the dose distribution, with weights determined by the first-stage compliance function (?).

The first-stage equation is:

$$\text{Dose}_i = \pi_0 + \pi_1 \text{Treatment}_i + X_i' \pi + u_i \quad (5)$$

and the second-stage equation is:

$$Y_{i,t} = \alpha + \beta \widehat{\text{Dose}_i} + X_i' \delta + \varepsilon_{i,t} \quad (6)$$

where $\widehat{\text{Dose}}_i$ denotes the predicted value from the first stage. For the extensive-margin specification, Dose_i is replaced by the binary Active_i indicator defined above, and β is the standard CACE. We report the first-stage F -statistic (Kleibergen-Paap for the continuous dose, conventional for the binary indicator) alongside each 2SLS estimate. For the continuous dose specification, if the first-stage F -statistic falls below the ? critical value for 10 percent maximal IV size ($F < 16.38$ for one endogenous regressor and one instrument), we will report Anderson-Rubin confidence sets, which are robust to weak instruments, alongside the 2SLS point estimates.

Exploratory: non-linear dose-response. To test whether the returns to engagement are concave — consistent with a diminishing-returns model in which the tool transmits a bounded stock of expert knowledge — we estimate an augmented specification:

$$Y_{i,t} = \alpha + \beta_1 \widehat{\text{Dose}}_i + \beta_2 \widehat{\text{Dose}}_i^2 + X_i' \delta + \varepsilon_{i,t} \quad (7)$$

where both $\widehat{\text{Dose}}_i$ and $\widehat{\text{Dose}}_i^2$ are instrumented by Treatment_i and $\text{Treatment}_i \times Z_i$, with Z_i a baseline covariate that shifts compliance intensity (we pre-specify career stage for this role, since junior faculty have more active project slots by design). A negative β_2 indicates concavity: diminishing returns to additional feedback cycles, consistent with convergence toward a quality ceiling as the researcher internalizes frontier standards. This specification is reported as exploratory because power for the quadratic term is limited at $N = 148$.

Exclusion restriction and threats. The exclusion restriction requires that random assignment affects the outcome only through engagement with the structured multi-agent tool. This assumption is plausible because the control group retains access to general-purpose LLMs throughout the study period, so the instrument isolates the specific effect of structured, multi-agent AI feedback rather than any AI use at all. We nonetheless acknowledge that assignment to the waitlist may induce the control group to increase their use of general-purpose LLMs as a substitute; we test for this directly by comparing self-reported

general-purpose LLM usage at endline across treatment and control groups (items A2 and A3), and interpret the LATE as a lower bound on the structured-feedback effect if differential adoption is detected.

Because compliance may differ systematically by career stage, we also report the Complier Average Causal Effect (CACE) separately by stratum, estimating the first-stage equation with the interaction $\text{Treatment}_i \times \text{JuniorFaculty}_i$ to allow the first-stage coefficient to vary across groups. Given that each stratum contains approximately $N = 37$ treated participants, and that the observed compliance rate of approximately 70 percent reduces the effective complier sample within each stratum to roughly $N \approx 26$, these stratum-specific estimates are underpowered for all but large effects ($\text{MDES} \approx 0.55 \text{ SD}$) and are therefore reported as exploratory. Their primary value is in characterizing whether compliance is differential by career stage, which would confound the heterogeneity analysis.

Robustness to the definition of engagement. We assess the sensitivity of the LATE to how engagement is measured across a pre-specified set of alternative definitions. On the *extensive margin* we vary the adoption threshold: (i) any logged platform activity versus no activity, (ii) at least one completed feedback cycle (the primary definition), and (iii) engagement above the sample median among active users. On the *intensive margin* we vary the underlying dose: (i) the number of distinct manuscripts processed, (ii) the cumulative number of completed feedback cycles (the primary definition), (iii) the number of distinct active project slots used, and (iv) the self-reported engagement item (A5, Q68), each standardized. For each definition we re-estimate the first stage and the 2SLS second stage, report the first-stage F -statistic (and Anderson-Rubin confidence sets where the first stage is weak), and assess whether the sign, statistical significance, and approximate magnitude of the treatment effect are sensitive to the choice of engagement measure.

6 Threats to Internal Validity

The collaborative nature of research production in academic economics raises two distinct SUTVA concerns, both of which the design addresses directly. First, direct contamination — treated researchers sharing credentials with control-group colleagues — is mitigated by the Active Project Slot architecture as described in Section 3.3. The opportunity cost of account sharing exceeds the benefit for any researcher with active job-market papers, providing an incentive-compatible safeguard. Horizontal contamination through shared collaboration — a treated and a control participant working together on the same paper — is detectable in our data: we collect the names of each participant’s most frequent coauthors at baseline, which lets us flag treatment–control coauthor links and assess the robustness of the estimates to their exclusion.

A more direct threat to the treatment contrast is the possibility that control-group participants take up a specialised research-feedback tool similar to ResearchAI during the study window, which would narrow the gap between the arms. General-purpose LLM use is not itself the concern, since it is available to both arms throughout. We measure this directly. At baseline we record whether each participant uses generic AI assistants or specialised AI tools tailored for academic research, and we ask the same question at the midline and endline waves.

6.1 Demand Characteristics and Response Inflation

Three features of our design address the standard demand-characteristics concern that applies to any field experiment using subjective outcomes.. First, all productivity items are framed as within-person flows with an explicit midpoint of 5 denoting “no change since the start of the study,” which anchors each respondent to her own baseline and reduces the scope for socially desirable upward drift. Second, the survey embeds an instructed-response attention check (item A4) that requires respondents to select “Rarely” regardless of their

true answer; we will drop respondents who fail this check from robustness specifications and report the share of failures by arm as a test for differential inattention. Third, and most directly, the intellectual ownership items (O1–O3) are scored so that higher concern about AI dependence reduces the construct of interest; a respondent inflating productivity reports would have an incentive to also underreport ownership concerns, and the joint pattern across productivity and ownership items therefore provides an internal consistency check on the direction of any demand-induced bias. The self-reported design is, moreover, the appropriate measurement strategy for this setting: perceived ease of producing high-quality work is the proximal margin on which an expert-feedback intervention acts, and it is precisely what objective output counts — uninformative in a six-month window for long-cycle knowledge production — cannot capture.

Tiene sentido añadr sección de contribución?

7 Administrative Information

7.1 Ethics and Institutional Review

The study has received ethics approval from the Institutional Review Boards of KU Leuven and the London School of Economics. All participants provided informed consent prior to completing the baseline survey, with a clear statement that their survey responses and platform usage data would be used for research purposes.

7.2 Declaration of Interest

The authors state that they have no competing financial or non-financial interests related to this study. The ResearchAI Development Suite was built by the research team for the purposes of this study; the authors do not have any commercial stake in the platform.

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A Survey Instrument and Variable Construction

Table A.1: Survey Instrument across Waves (Baseline, Midline, Endline)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
<i>Block 1. Contact information</i>					
1	1	What is your first name?	First name	Open-ended (text)	B
1	2	What is your surname?	Surname	Open-ended (text)	B
1	3	What is your current institution?	Institution	Open-ended (text)	B
1	4	What is your institutional email address?	Institutional email	Open-ended (text, email validation)	B
1	5	Please retype your institutional email address.	Email (confirm)	Open-ended (text, email validation)	B

Continued on next page

Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
1	6	What is your ORCID iD?	ORCID iD	Open-ended (text)	B
1	7	Link to your Google Scholar profile.	Google Scholar	Open-ended (text)	B
<i>Block 2. Demographics and background</i>					
2	8	Are you a full-time student/researcher?	Full-time status	Yes; No	B
2	9	Current academic position.	Academic position	PhD student; Post-doc; Assistant Professor; Associate Professor; Other	B
2	10	Year you started your PhD.	PhD start year	Open-ended (numeric)	B
2	11	Year you completed (or expect to complete) your PhD.	PhD completion year	Open-ended (numeric)	B
2	12	What is your gender?	Gender	Male; Female	B
2	13	Is English your mother tongue?	English native	Yes; No	B
2	14	Main field of study.	Field of study	Microeconomics; Macroeconomics; Econometrics; Labour or Public Economics; IO, Finance, Trade; Political Economy; Other	B
2	15	Do you have caring responsibilities?	Caring responsibilities	Yes; No	B

Continued on next page

Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
2	16	Did either parent/caregiver complete a university degree or higher?	Parental education	Yes; No; Don't know	B
<i>Block 3. Research productivity (primary outcome; flows since study start)</i>					
3	17	Ease of identifying and fixing the main weaknesses in your research design or empirical strategy.	PQ1	0–10, since study start: 0 = much harder, 5 = no change, 10 = much easier	M / E
3	18	Ease of working out and stating what your paper adds to the literature.	PQ2	0–10, since study start: 0 = much harder, 5 = no change, 10 = much easier	M / E
3	19	Ease of identifying the larger economic question and why it matters.	PQ3	0–10, since study start: 0 = much harder, 5 = no change, 10 = much easier	M / E
3	20	Overall ease of bringing your main paper up to target-journal quality. (<i>headline endpoint</i>)	PQ4	0–10, since study start: 0 = much harder, 5 = no change, 10 = much easier	M / E
3	21	For the same amount of work, overall quality of research you can produce.	PQ5	0–10, since study start: 0 = much lower, 5 = no change, 10 = much higher quality	M / E
3	22	Number of research papers or drafts you are actively working on.	PQ6	0–10, since study start: 0 = many fewer, 5 = same, 10 = many more	M / E
3	23	Pace of moving papers toward submission (getting drafts ready to send out).	PQ7	0–10, since study start (5 = no change)	M / E

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
<i>Block 4. Research output and pipeline</i>					
4	24	Papers published in peer-reviewed journals.	Journal publications	0; 1–3; 4–8; 9–15; 16 or more	B
4	25	Papers currently awaiting a journal decision.	Papers under review	Open-ended (numeric)	B
4	26	Papers currently being prepared for submission.	Papers in progress	Open-ended (numeric)	B
4	27	Published papers in AEA, JPE, Econometrica, QJE, ReStud, JOLE, or ReStat.	Top-7 publications	Open-ended (numeric)	B
4	28	Papers under review at those seven journals.	Top-7 under review	Open-ended (numeric)	B
4	29	Papers being prepared for submission within the next month.	Submissions next month	Open-ended (numeric)	B
4	30	Overall state of your main research project (since study start).	P3	5-pt ordinal: Much further along ... Much further behind	M / E
4	31	Journal submissions since study start (+ tier of most recent target).	P1	0; 1; 2; 3 or more (+ target tier)	E
4	32	New projects or draft chapters begun since study start.	P2	0; 1; 2; 3 or more	E

Block 5. Time allocation across research tasks (frequency; since study start at M/E)

Continued on next page

Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
5	33	Collecting, cleaning, or preparing data for analysis.	T1	1–5 Likert: Never; Rarely; Sometimes; Often; Very Often	B / M / E
5	34	Writing, debugging, or improving code; running analyses or simulations.	T2	1–5 Likert (as above)	B / M / E
5	35	Writing or revising research papers or drafts.	T3	1–5 Likert (as above)	B / M / E
5	36	Editing or improving clarity, structure, and presentation of written work.	T4	1–5 Likert (as above)	B / M / E
5	37	Generating research ideas or hypotheses.	T5	1–5 Likert (as above)	B / M / E
5	38	Reading and synthesising academic papers to inform your research.	T6	1–5 Likert (as above)	B / M / E
5	39	Developing or refining the conceptual framing or contribution of a project.	T7	1–5 Likert (as above)	B / M / E
5	40	Discussing research progress or problems with co-authors.	T8	1–5 Likert (as above)	B / M / E
5	41	Planning or coordinating tasks and timelines within a research project.	T9	1–5 Likert (as above)	B / M / E
5	42	Leading a funding application or grant proposal.	T10	1–5 Likert (as above)	B / M / E

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
5	43	Leading the overall direction of a research project.	T11	1–5 Likert (as above)	B / M / E
<i>Block 6. Senior colleagues, feedback and autonomy (flows since study start)</i>					
6	44	Reliance on senior colleagues for feedback on overall quality and direction.	K1	0–10, since study start: 0 = rely much more, 5 = same, 10 = rely much less	M / E
6	45	Ability to get useful feedback and move forward without waiting for a senior colleague.	K2	0–10, since study start: 0 = much less able, 5 = no change, 10 = much more able	M / E
6	46	Complexity of problems you can resolve on your own before escalating.	K3	0–10, since study start: 0 = much lower, 5 = no change, 10 = much higher	M / E
6	47	Closeness of your own assessment of weaknesses to a senior colleague's / referee's.	K4	0–10: 0 = not at all close, 5 = somewhat, 10 = extremely close	M / E
6	48	Frequency of seeking senior input on core design or direction.	K5	0–10, since study start: 0 = much more often, 5 = same, 10 = much less often	M / E
6	49	Number of active research collaborations.	K6	0–10, since study start: 0 = many fewer, 5 = same, 10 = many more	M / E
<i>Block 7. Research network and collaboration</i>					
7	50	Number of active research co-authors.	Active co-authors	Open-ended (numeric)	B

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
7	51	Active co-authors more junior than you.	Co-authors (junior)	Open-ended (numeric)	B
7	52	Active co-authors more senior than you.	Co-authors (senior)	Open-ended (numeric)	B
7	53	Name of the co-author you collaborate with most frequently.	Main co-author	Open-ended (text)	B
7	54	Name of the second most frequent co-author (if any).	Second co-author	Open-ended (text)	B
<i>Block 8. Intellectual ownership and AI dependence (flows since study start)</i>					
8	55	Confidence in using AI as a useful part of your workflow without losing ownership of ideas.	O1 [†]	0–10, since study start: 0 = less confident / more dependent, 5 = no change, 10 = more confident / in control	M / E
8	56	Sense that you are genuinely learning skills rather than outsourcing judgment.	O2 [†]	0–10, since study start: 0 = more outsourcing, 5 = no change, 10 = more learning	M / E
8	57	Concern that AI use might reduce your ability to independently evaluate your research.	O3	0–10, since study start: 0 = much less concerned, 5 = no change, 10 = much more concerned (reverse-coded)	M / E
<i>Block 9. Working life and research environment</i>					
9	58	Overall satisfaction with your work–life balance.	W1	1–5: Very dissatisfied ... Very satisfied	B / M / E

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
9	59	Frequency of uninterrupted blocks of at least 2–3 hours for research.	Research time blocks	1–5 Likert: Never; Rarely; Sometimes; Often; Very often	B
9	60	Change in frequency of uninterrupted 2–3 hour research blocks (flow).	W2	0–10, since study start: 0 = much less often, 5 = same, 10 = much more often	M / E
9	61	Access to people who help you navigate research and academic careers.	W3	1–5: Strongly disagree ... Strongly agree	B / M / E
<i>Block 10. Use of AI tools and study experience</i>					
10	62	Do you currently use AI tools in your research workflow?	AI use (any)	Yes; No	B
10	63	Type of AI tools used in research.	AI tool type	Generic AI assistant (e.g., ChatGPT, Claude, Gemini); Specialised AI tools for academic/economic research; I do not use AI tools	B
10	64	How you mainly use AI tools.	AI use mode	MCQ, multi-select: Ideation & feedback; Writing; Background research; Data analysis; Result interpretation; Coding; Mathematical derivations; I don't use AI	B

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
10	65	Confidence in using AI tools effectively in research.	AI confidence	Scale 0 (not at all) – 10 (very confident)	B
10	66	Frequency of general-purpose AI assistant use (since study start).	A2	1–5 Likert: Never . . . Very Often	M / E
10	67	Frequency of specialised research-AI use, excluding any study-provided tool (since study start).	A3	1–5 Likert: Never . . . Very Often	M / E
10	68	Extent to which you acted on ResearchAI feedback when revising your draft.	A5 [†]	0–10: 0 = did not act on it, 10 = substantially changed the paper	M / E
10	69	Importance to you that ResearchAI is shown to be effective for researchers like you.	A6 [†]	0–10: 0 = not at all important, 10 = extremely important	M / E
10	70	Apart from ResearchAI, which other AI tools have you used in your research since the study began?	A7 [†]	Multi-select: A general-purpose assistant (e.g., ChatGPT, Claude, Gemini); Another specialised research/economics AI tool; None	M / E

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Table A.1 (continued)

Block	Q#	Question wording	Item / code	Answer values/scale	Wave
<i>Notes:</i> Wave: B = Baseline (month 0), M = Midline (month 3), E = Endline (month 6). Blocks 1–2 are administered at baseline only. Items coded T1–T11, W1, and W3 are asked at baseline (recall window “past two weeks”) and at follow-up (“since the beginning of the study”); follow-up flow items anchored at 5 are differenced against their baseline value. [†] Administered to the treatment arm only (the control questionnaire omits ResearchAI references, sees only ownership item O3, and omits A5–A7). Variable construction and item–code mapping are given in Table A.2.					

Table A.2: Variable Construction Table

Category	Variable	Label / description	Source	Construction rule	Wave
A. Output					
Output	PQ1 Credibility, effort	Ease of identifying and fixing main weaknesses in design / empirical strategy	Survey	PQ1, Q17 (0–10 flow, 5 = no change); FDR-controlled primary family	M, E
Output	PQ2 Originality, effort	Ease of stating what the paper adds to the literature	Survey	PQ2, Q18; FDR-controlled primary family	M, E
Output	PQ3 Significance, effort	Ease of identifying the larger economic question and why it matters	Survey	PQ3, Q19; FDR-controlled primary family	M, E
Output	PQ4 Overall effort to target quality (<i>headline</i>)	Overall ease of bringing the main paper to target-journal quality	Survey	PQ4, Q20; headline confirmatory endpoint	M, E
Output	PQ5 Quality for given effort	For the same work, overall quality attainable	Survey	PQ5, Q21; FDR-controlled primary family	M, E
Output	PQ6 Active papers, quantity	Number of papers/drafts actively worked on	Survey	PQ6, Q22; differenced against baseline in_progress (Q26)	M, E

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Table A.2 (continued)

Category	Variable	Label / description	Source	Construction rule	Wave
Output	PQ7 Submission pace	Pace of moving drafts toward submission	Survey	PQ7, Q23	M, E
Output	Submission attempts (P1)	Journal submissions since study start (+ target tier)	Survey	P1, Q31; count since start	E
Output	New drafts initiated (P2)	New projects / draft chapters begun since start	Survey	P2, Q32; count since start	E
Output	Project progress (P3)	Self-assessed progress of main project since start	Survey	P3, Q30; 5-pt ordinal change	M, E
Output	Working-paper circulation	Distinct working papers circulated over the study window	Admin	Count of new working-paper records extracted from ORCID / Google Scholar over months 0–6	E
B. Mechanism					
Mechanism	High-context task frequency (E1–E5)	Frequency of judgment-intensive tasks requiring expert knowledge	Survey	Five items, each follow-up minus baseline: E1 ideas (Q37), E2 framing (Q39), E3 coauthor discussion (Q40), E4 leading direction (Q43), E5 planning (Q41); tested individually	B, M, E

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Table A.2 (continued)

Category	Variable	Label / description	Source	Construction rule	Wave
Mechanism	Low-context task frequency (R1–R5)	Frequency of codifiable tasks AI substitutes for most readily	Survey	Five items, each follow-up minus baseline: R1 data prep (Q33), R2 code/analysis (Q34), R3 writing/revising (Q35), R4 editing clarity (Q36), R5 reading/synthesis (Q38); key test is the high–low contrast	B, M, E
Mechanism	Knowledge-frontier items (K1–K6)	Reliance on / autonomy from senior colleagues and collaboration	Survey	K1–K6, Q44–Q49 (0–10 flow); tested individually (Garicano channel)	M, E
Mechanism	Intellectual ownership guardrail (O1–O3)	Retained ownership of ideas vs. outsourced judgment	Survey	O1 [†] , Q55; O2 [†] , Q56; O3 (reverse-coded), Q57; secondary guardrail, reported item by item	M, E
Mechanism	Usage intensity	Engagement breadth on the platform	Log	Unique manuscript uploads per month, months 1–6	M, E
Mechanism	Iterations per manuscript	Depth of iterative engagement	Log	Count of upload–revise cycles (incl. version comparisons) per manuscript	M, E

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Table A.2 (continued)

Category	Variable	Label / description	Source	Construction rule	Wave
Mechanism	Active project slots	Breadth of platform use across projects	Log	Number of distinct active project slots used	M, E
Mechanism	Adoption / dose (Active _{<i>i</i>})	Compliance measure instrumented by assignment in the LATE	Log	Extensive: ≥ 1 completed feedback cycle; intensive: standardised dose (uploads, iterations, slots), with self-reported A5, Q68 as companion	M, E
C. Control					
Control	Career stage (stratum, moderator)	PhD student vs. postdoc / junior faculty	Survey	From position , Q9; stratification fixed effect and component of the expertise moderator M_i	B
Control	Baseline mentoring access (moderator)	Access to people who help navigate research/careers	Survey	Baseline value of W3, Q61 (1–5 agreement); component of M_i	B
Control	Institutional tier (moderator)	Density of the expert environment	Survey	Derived from institution , Q3; component of M_i	B

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Table A.2 (continued)

Category	Variable	Label / description	Source	Construction rule	Wave
Control	Baseline covariate vector (X_i)	Demographics and publication profile entering all specifications	Survey	Gender (Q12), English native (Q13), field (Q14), caring (Q15), parental education (Q16); <code>pubs_total</code> (Q24), <code>under_review</code> (Q25), <code>in_progress</code> (Q26), <code>pubs_top7</code> (Q27), <code>paper_nextmonth</code> (Q29); baseline AI usage (Q62–Q65)	B
Control	Baseline task-frequency levels (flow anchors)	Recall-window task frequencies used to difference flow mechanisms	Survey	T1–T11, Q33–Q43 at baseline; subtracted from follow-up to form E/R flows	B
D. Spillover					
Spillover	Treatment–control coauthor link	Shared-collaboration spillover between arms (SUTVA)	Survey	Flag = 1 if a participant’s named coauthors (<code>coauthor_name_1</code> , Q53; <code>coauthor_name_2</code> , Q54) match a participant in the opposite arm; estimates re-checked excluding linked pairs	B
E. Contamination					

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Table A.2 (continued)

Category	Variable	Label / description	Source	Construction rule	Wave
Contamination	Specialised research-AI uptake (control)	Control-arm take-up of a substitute research-feedback tool	Survey	A3, Q67 at follow-up (use of specialised research AI excluding the study tool), benchmarked to baseline <code>ai_use</code> (Q62) / <code>ai platform</code> (Q63)	M, E
Contamination	General-purpose LLM use	Substitution toward generic assistants (waitlist substitution)	Survey	A2, Q66 at follow-up; compared across arms to bound the LATE	M, E
Contamination	Other AI tools used (treatment)	Within-treatment use of non-study AI tools	Survey	A7 [†] , Q70 (multi-select)	M, E

Notes: Q# refer to the question numbers in Table A.1. Source: Survey = participant survey; Log = ResearchAI platform server logs; Admin = bibliographic records linked from ORCID/Google Scholar with consent. Wave: B = baseline (month 0), M = midline (month 3), E = endline (month 6). All 0–10 “flow” items are reported by the respondent as a within-person change since study start, with 5 = no change. Task-frequency items (1–5 Likert) are asked at baseline (recall window) and at follow-up, and differenced against the baseline value. Productivity, mechanism, and ownership items are tested individually with the Anderson (2008) FDR correction; no composite index is formed. [†] Treatment arm only; the control instrument omits these items.